

Supplementary Information

Geographic and Linguistic Bias in LLMs for Air Pollution Policy: A 25-Country, 14-Model, 4-Language, 5-Task Benchmark

Reproducibility. Every data-driven table below (Tables [S1–S7](#)) is regenerated from the raw confirmatory artifacts by `code/analysis/make_supplement_tables.py`; no value is hand-entered. The primary judge for all reported composites is `gpt-5-mini-2025-08-07`. The analysed confirmatory corpus comprises 9,251 judge-scored responses (7,580 English-prompt; 1,671 native-language) over 25 countries and 14 deployment stacks. All inputs, the scoring code, the ground-truth registry, and the anonymised responses will be released on publication under CC-BY-4.0 (data/prompts) and MIT (code).

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S1 Pre-registration, design, and country selection

The hypotheses, primary models, persona protocol, exclusion criteria, multiple-comparison corrections, and the country sampling frame were specified in advance of confirmatory data collection. The pre-registered design covers **15 countries**; the sample was subsequently extended, *post*

registration, to **25 countries** to gain power for the development gradient (H1) and the corpus-mechanism tests (H4) and to multiply the within-country language pairs (H2). Throughout the manuscript and this supplement, every effect computed on the 25-country sample is reported alongside its pre-registered 15-country value, and no pre-registered conclusion is revised on the basis of the extension alone (Section S8).

Countries were selected by theoretical sampling triangulated across three independent classifications: the UNCTAD Global North/South classification (operationalising the coloniality-of-knowledge framing), the Joshi et al. (2020) six-class linguistic-resource taxonomy [1], and World Bank income groups. The pre-registered 15 (Set “P” in Table S4) span four world regions, five Joshi classes, and four income groups; the 10 extension countries (Set “E”) comprise seven additional Global North references (United Kingdom, Canada, Australia, South Korea, France, Italy, Portugal) chosen to thicken the high-development end of the gradient, and three additional Global South native-language-pair countries (Colombia, Chile, Angola). The full design rationale is in docs/supplement/s1_design_rationale.md.

S2 Deployment-stack roster and serving configuration

Unit of analysis. We evaluate each model *as served* through a fixed access stack — the tuple of model weights, vendor, serving infrastructure, and API/endpoint layer — rather than idealized weights in isolation, because a public manager interacts with the served stack and any measured gap is a property of that stack. Table S1 lists the 14 stacks with their exact API/tag strings and execution venues.

Table S1. Deployment-stack roster. Each model is evaluated *as served* through a fixed access stack; the API/tag string and execution venue are part of what each label denotes. Parameter counts are vendor-reported (dense or total for MoE; closed-frontier counts undisclosed). *N* and mean are realized English-prompt confirmatory coverage.

Model	Vendor	Tier	API / tag string	Venue	<i>N</i>	Mean
DeepSeek-V3	DeepSeek	A	deepseek-chat	DeepSeek API	500	0.625
Command R+	Cohere	A	command-r-plus-08-2024	Cohere (trial)	500	0.509
Llama 3.3 70B	Meta	A	llama-3.3-70b-versatile	Groq (free)	474	0.506
GPT-OSS 120B	OpenAI	A	openai/gpt-oss-120b	Groq (free)	298	0.485
Llama 4 Scout	Meta	A	meta-llama/llama-4-scout-17b-16e-instruct	Groq (free)	500	0.473
Qwen3 32B	Alibaba	B	qwen/qwen3-32b	Groq (free)	498	0.546
Phi-4 14B	Microsoft	B	phi4	Ollama (local)	500	0.485
Qwen3 14B	Alibaba	B	qwen3:14b	Ollama (local)	500	0.474
Llama 3.1 8B	Meta	C	llama-3.1-8b-instant	Groq (free)	777	0.424
Cabra-Mistral 7B	botbot-ai	C	cabra-mistral-7b	Ollama (local)	496	0.320
GPT-5-mini	OpenAI	D	gpt-5-mini	OpenAI API	777	0.626
Gemini 2.5 Flash	Google	D	gemini-2.5-flash	Google AI (free)	490	0.604

Model		Vendor	Tier	API / tag string	Venue	N	Mean
Claude 4.5	Haiku	Anthropic	D	claude-haiku-4-5	Anthropic API	490	0.597
GPT-5		OpenAI	E	gpt-5	OpenAI API	780	0.647

Determinism and seed handling. All stacks were queried at temperature 0.3 with the maximum-determinism configuration each provider documents, not an identical parameter surface, because the surfaces differ. Seed support is heterogeneous: locally served stacks (Ollama) honour a fixed seed; OpenAI exposes an advisory seed; Anthropic’s Messages API exposes no seed parameter; and for the remaining hosted endpoints a seed is either advisory or not exposed. For every call we record the requested seed and a `seed_status` field (`sent`, `logged-only`, or `not-supported`) in the released run metadata, so that the determinism actually achieved is auditable rather than assumed. Two stochastic replications per (prompt, model, persona) cell estimate within-cell variability.

Acquisition window. Served-model behaviour can drift as vendors update infrastructure, so the reported accuracies are point-in-time estimates relative to the collection window. The pinned tag, version string, and acquisition timestamp are recorded for every call. Confirmatory collection ran from **2026-04-26 to 2026-06-12** (UTC), spanning the pilot, the pre-registered 15-country collection, and the post-registration extension to 25 countries. Table S2 gives the first and last non-error call timestamp per provider together with the official API documentation consulted; the full per-call timestamps are released with the dataset.

Table S2. Per-provider acquisition window and official API documentation. Reported accuracies are point-in-time estimates relative to this window; first/last are the earliest and latest non-error call timestamps (UTC) recorded in the run logs. The window spans the pilot, the pre-registered 15-country collection, and the post-registration extension.

Provider (venue)	First (UTC)	Last (UTC)	API documentation
Anthropic API	2026-04-27	2026-06-12	https://docs.anthropic.com/en/api/messages
Cohere (trial)	2026-04-26	2026-06-11	https://docs.cohere.com/reference/chat
DeepSeek API	2026-04-26	2026-06-11	https://api-docs.deepseek.com
Google AI (free)	2026-04-27	2026-06-11	https://ai.google.dev/api
Groq (free)	2026-04-26	2026-06-11	https://console.groq.com/docs/api-reference
Ollama (local)	2026-04-27	2026-06-12	https://github.com/ollama/ollama/blob/main/docs/api.md

Provider (venue)	First (UTC)	Last (UTC)	API documentation
OpenAI API	2026-04-27	2026-06-12	https://platform.openai.com/docs/api-reference/chat
All providers	2026-04-26	2026-06-12	—

S3 Coverage

Per-cell coverage is not uniform across the roster: free and rate-limited endpoints returned fewer admissible responses than the metered APIs, and one candidate (Gemini 2.5 Flash on early native batches) and a sixth judge (Command R+) hit provider quotas. We report realized coverage rather than silently dropping under-covered cells. Table S3 gives the model $\times$ task English-prompt matrix (column totals equal the by-task counts in the main text); Table S4 gives per-country English, native, and total counts with tier and pre-registration set; Table S5 lists the native-language pairs used for H2. The complete model $\times$ country $\times$ task coverage CSV, with the quota or exclusion reason for every shortfall, is included in the release.

Table S3. Coverage matrix: realized English-prompt confirmatory responses per model $\times$ task (T1 standard, T2 local datum, T3 health synthesis, T4 instruments, T5 recommendation). No cell is silently dropped; shortfalls are quota-driven (Methods).

Model	T1	T2	T3	T4	T5	Total
GPT-5	156	156	156	156	156	780
GPT-5-mini	156	156	156	153	156	777
Llama 3.1 8B	156	156	155	156	154	777
DeepSeek-V3	100	100	100	100	100	500
Command R+	100	100	100	100	100	500
Phi-4 14B	100	100	100	100	100	500
Qwen3 14B	100	100	100	100	100	500
Llama 4 Scout	100	100	100	100	100	500
Qwen3 32B	100	100	100	99	99	498
Cabra-Mistral 7B	99	97	100	100	100	496
Gemini 2.5 Flash	98	98	98	98	98	490
Claude Haiku 4.5	98	98	98	98	98	490
Llama 3.3 70B	98	97	96	92	91	474
GPT-OSS 120B	68	72	60	42	56	298
Total	1529	1530	1519	1494	1508	7580

Table S4. Coverage by country: tier (GN/GS), pre-registration status (P = pre-registered 15; E = post-registration extension), and realized confirmatory responses (English, native-language, total).

Country	Tier	Set	English	Native	Total
Colombia	GS	E	260	251	511
Chile	GS	E	260	241	501
Portugal	GN	E	261	240	501
Angola	GS	E	260	240	500
Argentina	GS	P	355	139	494
India	GS	P	352	138	490
Mexico	GS	P	329	140	469
Peru	GS	P	316	141	457
Brazil	GS	P	260	141	401
Germany	GN	P	356	0	356
Indonesia	GS	P	356	0	356
Bangladesh	GS	P	355	0	355
Egypt	GS	P	355	0	355
Kenya	GS	P	348	0	348
Japan	GN	P	347	0	347
Nigeria	GS	P	317	0	317
Philippines	GS	P	316	0	316
United States	GN	P	312	0	312
South Africa	GS	P	302	0	302
United Kingdom	GN	E	265	0	265
Canada	GN	E	260	0	260
Australia	GN	E	260	0	260
France	GN	E	260	0	260
Italy	GN	E	260	0	260
South Korea	GN	E	258	0	258
Total (25)			7580	1671	9251

Table S5. Native-language confirmatory responses by language and country (H2 within-country English/native contrast).

Native language (Joshi)	Country	<i>N</i> (native)
Portuguese (4)	Portugal	240
	Angola	240
	Brazil	141
Spanish (5)	Colombia	251
	Chile	241
	Peru	141
	Mexico	140

Native language (Joshi)	Country	N (native)
	Argentina	139
Hindi (4)	India	138

S4 Country covariates

Table S6 reports the developmental and corpus-representation covariates used in the mechanism analysis (H4). **Language-corpus** measures (Joshi class; mC4/OSCAR sizes reported in the main text) index how much text exists *in* a country’s language and are the candidate channel for the native-language penalty (H2). **Country-coverage** measures (English-Wikipedia article byte length, Wikidata sitelinks, Wikidata statements) index how broadly the encyclopedic corpus is *about* a country, independent of language, and are the candidate channel for the geographic gap (H1/H4). Because the geographic gap is measured *in English*, a residual North/South gap cannot be a language-corpus effect, which is why H4 isolates country coverage for H1 and language-corpus size for H2. All Wikidata entities are identified by a resolvable QID and were retrieved within the collection window.

Table S6. Country covariates. HDI: UNDP HDR 2023–24 (2022 data). Joshi: linguistic-resource class of the dominant official language [1]. Corpus measures from Wikimedia/Wikidata (entity QID resolvable): English-article byte length, Wikidata sitelinks (language editions), and Wikidata statements.

Country	Tier	HDI	Joshi	EN-wiki bytes	Sitelinks	WD stmts
Germany	GN	0.950	5	207,810	403	1402
Australia	GN	0.946	5	233,095	395	1674
United Kingdom	GN	0.940	5	368,888	391	1085
Canada	GN	0.935	5	281,016	393	1346
South Korea	GN	0.929	4	275,291	350	746
United States	GN	0.927	5	383,914	424	1710
Japan	GN	0.920	5	205,844	414	995
France	GN	0.910	5	279,498	413	1196
Italy	GN	0.906	4	305,234	405	1039
Portugal	GN	0.874	4	324,038	369	751
Chile	GS	0.860	5	232,108	379	689
Argentina	GS	0.849	5	267,662	369	881
Mexico	GS	0.781	5	263,757	398	1286
Peru	GS	0.762	5	239,544	332	867
Brazil	GS	0.760	4	301,844	384	1151
Colombia	GS	0.758	5	303,401	363	774
Egypt	GS	0.728	5	284,547	387	757
South Africa	GS	0.717	5	270,861	365	750
Indonesia	GS	0.713	3	226,349	362	1005
Philippines	GS	0.710	5	493,981	331	963

Country	Tier	HDI	Joshi	EN-wiki bytes	Sitelinks	WD stmts
Bangladesh	GS	0.670	3	275,786	343	814
India	GS	0.644	5	310,681	399	1679
Kenya	GS	0.601	5	216,189	334	699
Angola	GS	0.591	4	180,098	322	676
Nigeria	GS	0.548	5	266,041	349	1194

S5 Ground-truth registry (T1)

The validity of the geographic gradient (H1) requires that ground truth be equally available and equally verifiable across countries. For T1 (the binding annual PM_{2.5} standard) we anchored every country to an official primary source by documentary audit: the source text was retrieved, the value was read verbatim, and the entry records the value, status, source-of-record, a resolvable URL, and the retrieval method (Table S7). Three countries have *no* national PM_{2.5} standard, confirmed from the official text (Nigeria’s NESREA regulates PM₁₀ only; Argentina has no binding federal value; Angola has no national air-quality legislation); in those cases a faithful model must decline to state a non-existent threshold rather than fabricate one. This provenance is more reproducible than expert assertion: any reader can re-verify each value from the cited official source. No reference in the registry is unverified, and no value was carried over from secondary sources without confirmation against the primary text.

Table S7. Ground-truth registry of official primary sources by document type. VERIFIED_NO_STANDARD = verbatim excerpts are in the relea

Country	Value
Angola	NO national air-quality legislation / standard
Argentina	NO single binding federal PM2.5 standard (Ley 20.284/1973 framework; PM2.5 limits set
Australia	8 ug/m3 (annual; 7 target 2025)
Bangladesh	15 ug/m3 (annual) per Gazette S.R.O. 220-Law/2005; a 2022 revision exists but the revis
Brazil	17 ug/m3 (annual, PI-2)
Canada	8.8 ug/m3 (annual, CAAQS 2020)
Chile	20 ug/m3 (annual)

Country	Value
Colombia	25 ug/m3 (annual, in force since 2018; 15 from 2030; 24h=37)
Egypt	PM2.5 limit set under Law 4/1994 (amended 9/2009, 105/2015) Executive Regulations (o
France	25 ug/m3 (annual, in force; 10 from 2030)
Germany	25 ug/m3 (annual, in force 2026; 10 from 2030)
India	40 ug/m3 (annual)
Indonesia	15 ug/m3 (annual)
Italy	25 ug/m3 (annual, in force; 10 from 2030)
Japan	15 ug/m3 (annual)
Kenya	35 ug/m3 (annual; 24h 75)
Mexico	10 ug/m3 (annual)
Nigeria	NO national PM2.5 standard (PM10 annual=60 ug/m3, 24h=150)
Peru	25 ug/m3 (annual)
Philippines	25 ug/m3 (annual, 1-year guideline)

Country	Value
Portugal	25 ug/m3 (annual, in force; 10 from 2030)
South Africa	20 ug/m3 (annual)
South Korea	15 ug/m3 (annual)
United Kingdom	10 ug/m3 (annual, target by 2040)
United States	9.0 ug/m3 (annual, primary NAAQS, 2024)

S6 Composite outcome and scoring rubric

Every response was scored by the primary judge against the ground-truth registry entry (T1, T2, T4) or the validated rubric (T3, T5), producing five subcomponents each normalised to $[0, 1]$:

1. **Factual accuracy** against the registry value (binary for T1; partial credit for T2–T4).
2. **Contextual completeness** scored against a pre-validated rubric (T3, T4).
3. **Citation quality**, verified via DOI/official-source check.
4. **Absence of hallucination**, coded binarily (0 hallucinated, 1 faithful).
5. **Applied validity**, the rubric score for the operational recommendation (T5).

The primary composite uses a pre-specified weighting (factual 0.30, contextual 0.25, citation 0.15, calibration 0.15, hallucination 0.15). As a sensitivity analysis the headline effects are recomputed under two alternative weightings — equal weights and a PCA-derived set (the first principal component of the five subcomponents) — and a conclusion is treated as robust only if its direction holds under all three; all headline effects are stable (tier gap +4.9 to +6.6 pp; HDI gradient $\rho = 0.51$ – 0.53 ; factual floor δ from -0.59 to -0.64 ; regional-model deficit δ from -0.50 to -0.52). The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test. Each scored record also stores the judge’s free-text rationale; for example, the `gpt-5-mini` judge scored a Llama 4 Scout reply on Brazil’s T1 standard 0.15 with the rationale that it “gives wrong regulation (491/2018 vs correct 506/2024) and wrong current value... included a specific but incorrect citation and fabricated timing,” illustrating how the factual-accuracy and hallucination subcomponents discriminate substantive failure.

S7 Judge-panel reliability

To address the judge–target overlap (the primary judge, `gpt-5-mini`, is itself among the evaluated tier), a fixed stratified reliability sample of **131 responses** (stratified by country $\times$ task $\times$ persona, fixed seed) was independently re-scored by a vendor-diverse panel. The sample was frozen *before* judging — the agreement sampler otherwise re-draws from the live response base, which desynchronised the per-judge samples when the base grew from 15 to 25 countries — and pinned to the judges’ common items by `code/analysis/judge_fixed_keys.py`.

Table S8. Inter-judge reliability on the fixed 131-item sample. The operative quantity is the reliability of the panel *mean*, ICC(2, k); by the Spearman–Brown relation it is high even when single judges agree only moderately. Source: `judge_panel_reliability.json`.

Quantity	Value
Judges (k)	4 (OpenAI, Anthropic, Google, DeepSeek)
Common items	131
Krippendorff’s α (interval)	0.667
Single-judge ICC(2, 1)	0.672
Panel-mean ICC(2, 4)	0.891
Pairwise Pearson: GPT-5-mini Claude Sonnet	0.859
GPT-5-mini DeepSeek-V3	0.834
Claude Sonnet DeepSeek-V3	0.821
GPT-5-mini Gemini 2.5 Pro	0.626
Claude Sonnet Gemini 2.5 Pro	0.630
DeepSeek-V3 Gemini 2.5 Pro	0.610

Agreement is strongest among GPT-5-mini, Claude, and DeepSeek (Pearson 0.82–0.86) and lowest for the systematically more lenient Gemini (0.61–0.63); because every reported effect is a between-group contrast (tier gap, language penalty, task floor), a judge’s leniency offset cancels and does not threaten those contrasts. A fifth judge (Llama 3.3 70B, Meta) corroborated on the items it covered (five-judge ICC(2, 5) = 0.861) but, being a noisier 70B judge, *lowered* the panel α to 0.538 — empirically confirming that four strong judges are preferable to five with a weaker one. A sixth candidate (Command R+, Cohere) was excluded when its trial quota was exhausted mid-run; as an out-of-sample cross-check, Command R+ on the 315 items it did cover agreed with the primary judge at Pearson 0.647 / ICC(2, 1) = 0.602. The study uses the single `gpt-5-mini` composite for all effects, validated by this panel, rather than a judge-averaged score; the absence of an independent human-gold layer is stated as a limitation in the main text.

S8 Statistical methods and full confirmatory test outputs

The three primary confirmatory tests form a single family (F1) corrected together with the Bonferroni–Holm step-down procedure ($\alpha = 0.05$). Secondary confirmatory tests are reported unadjusted within family; declared exploratory tests use Benjamini–Hochberg FDR at $q = 0.10$. Rank correlations use Spearman ρ ; monotonic-trend robustness uses the Mann–Kendall test; the persona effect uses a country-level permutation test (5,000 permutations). All outputs below are regenerated by `code/analysis/formal_tests.py`, `robust_tests.py`, and `h4_corpus_mechanism.py`.

Table S9. Primary family (Bonferroni–Holm): pre-registered 15-country values beside the post-registration 25-country extension. H1-by-HDI is the one primary test that rejects its null at $n = 25$; it remains below the pre-registered $\rho \geq 0.55$ effect-size threshold in both samples.

Test	15 countries (pre-registered)	25 countries (extension)
H1 Spearman $\rho(\text{acc}, \text{HDI})$	+0.143 (one-sided $p = 0.306$), ns	+0.512 ($p = 0.004$), rejects under B–H
H1 Mann–Kendall (by HDI)	$S = 9$, $Z = +0.40$, $p = 0.692$	$S = 102$, $Z = +2.36$, $p = 0.018$
H1 Spearman $\rho(\text{acc}, \text{Joshi})$	+0.028, ns	−0.061, ns (development, not linguistic)
H1 vs threshold $\rho \geq 0.55$	not met	not met (significant, below effect size)
H4 partial $\rho(\text{acc}, \text{Wiki} \text{HDI})$	+0.218 ($p = 0.45$), ns	+0.036 ($p = 0.87$), ns
H6 DiD persona (GS–GN)	+0.97 pp, perm $p = 0.142$, ns	+0.39 pp, perm $p = 0.257$, ns

Table S10. H4 corpus mechanism, multi-measure (25 countries). Country-coverage breadth (Wikidata sitelinks) predicts accuracy at the zero-order level and survives multiplicity correction across the three coverage proxies, but attenuates to non-significance after adjusting for development (HDI); we therefore report it as suggestive/exploratory, not as an established mechanism. The pre-registered language-edition proxy is null.

Measure	Family	$\rho(\text{acc}, \text{log measure})$	partial HDI
English-Wikipedia bytes	country	+0.198 ($p = 0.34$)	+0.181
Wikidata sitelinks	country	+0.539 ($p = 0.005$)	+0.317 ($p = 0.13$)
Wikidata statements	country	+0.313 ($p = 0.13$)	+0.144
Wikipedia-edition count	language	+0.082, ns	+0.036, ns

S9 Additional robustness and corroboration analyses

The headline effects were re-tested under several pre-specified analyses, all regenerated by `code/analysis/weight_glm_and_manipcheck.py`, `bayesian_reestimation.py`, and `mediation_h4.py` (wired into `run_all.py --conf`

Composite-weighting sensitivity. Recomputing every effect under the primary author-specified weighting, equal weights, and a PCA-derived weighting (first principal component of the five sub-components) leaves all four headline conclusions stable in direction (Table S12).

Mixed model and Bayesian re-estimation. A Gaussian linear mixed model (`statsmodels` MixedLM, REML) with crossed random intercepts for country and model gives a Global South effect of $\beta = -0.077$ (SE = 0.028, $p = 0.007$; country-only -0.077 , model-only -0.064). A Bayesian hierarchical re-estimation with weakly informative priors (`pymc`; crossed country/model intercepts) agrees: posterior mean -0.065 , 94% HDI $[-0.096, -0.033]$, $P(\beta < 0) = 1.00$ ($\hat{R} = 1.01$).

E-values. The unmeasured confounding required to explain away the supported geographic effects is substantial: E-value = 1.85 for the tier gap and 5.37 for the development gradient (VanderWeele & Ding 2017 approximation from the standardized effect).

Exploratory mediation (H4). A standardized country-level path model (`semopy`, $n = 25$): HDI $\rightarrow$ sitelinks $a = +0.67$; sitelinks $\rightarrow$ accuracy $b = +0.27$; direct $c' = +0.31$; indirect $ab = +0.18$ with bootstrap 95% CI $[-0.27, +0.59]$ (includes zero). Country coverage is thus a suggestive but not established mediator, consistent with the main-text framing.

Table S11. Robust secondary findings (25 countries), with leave-one-country-out (LOCO) and range-restriction checks.

Finding	Statistic
Tier gap (GN–GS)	+6.2 pp, 95% CI [+3.7, +8.6] (15-country: +6.7 [+2.8, +10.2])
Tier gap, LOCO range	[+5.8, +7.1] pp, stable
H1 HDI gradient, LOCO range	$\rho \in [+0.48$ (drop NGA), +0.66 (drop IND)]
H1 range-restricted (within 15-country HDI span)	$\rho = 0.512$ ($p = 0.009$): density, not range
Task floor (T1+T2 vs T3–T5)	0.367 vs 0.638, Mann–Whitney, Cliff’s $\delta = -0.64$
H3 Cabra–Mistral vs rest	0.320 vs 0.543, Cliff’s $\delta = -0.51$
H2 native vs English (all)	$\Delta = -2.1$ pp, Wilcoxon $p = 1.96 \times 10^{-3}$ ($n = 762$)
Spanish / Portuguese / Hindi	-1.0 ($p = 0.15$) / -2.3 ($p = 0.033$) / -7.8 ($p = 0.013$)
H2 country-level (independent unit, $n = 9$)	9/9 negative, mean -2.4 pp, Wilcoxon $p = 0.008$
H5 open vs closed-accessible	+14.1 pp closed advantage (descriptive)

Table S12. Headline effects under three composite weightings (English confirmatory, $n = 7,580$). Direction is stable throughout.

Weighting	Tier gap (pp)	ρ (HDI)	Floor δ	H3 δ
Author-specified (primary)	+6.24	+0.512	−0.638	−0.508
Equal	+4.85	+0.530	−0.588	−0.524
PCA-derived	+6.59	+0.517	−0.642	−0.496

Persona manipulation check. Across 4,349 persona-condition responses, 50.2% explicitly acknowledge the public-manager role frame, so the null persona effect (H6) is not an artifact of the frame being universally ignored.

S10 Prompt construction, persona template, and verbatim examples

Prompts were authored in English for the air pollution policy domain with domain supervision (PPGSAU/UTFPR) and validated against the ground-truth registry before entering the confirmatory stimulus set; a pilot subset met an inter-rater threshold of Krippendorff’s $\alpha \geq 0.70$ before scaling. Native-language renderings (Portuguese, Spanish, Hindi) were produced by a high-quality LLM translator (Claude Sonnet 4.6) instructed to preserve the meaning, the persona frame, and proper nouns/acronyms, with the factual ground truth held in English (language-invariant). Human native-speaker translation and back-translation verification were not performed, and this is noted as a limitation in the main text. Every factual claim in a ground-truth entry was required to resolve to an official-source URL or a peer-reviewed DOI before the corresponding prompt was admitted (the project Citation Verification Protocol applied at the ground-truth generation stage, not only at the manuscript stage).

Persona manipulation. Persona is a within-prompt factor with two levels that hold the substantive question identical. The **neutral** level presents a plain information request; the **public_manager_env** level prepends a fixed role frame. The contrast is shown verbatim for Brazil, T1:

Neutral. “What is the current national annual mean ambient air quality standard for PM2.5

in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”

Public environmental manager. “*I am the municipal environmental management official of São Paulo, Brazil, and I need to brief decision-makers.* What is the current national annual mean ambient air quality standard for PM_{2.5} in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”

The corresponding registry ground truth is: “Brazil’s national PM_{2.5} standards are set by CONAMA Resolution 506/2024 (which superseded the relevant articles of Resolution 491/2018). The stage in force in 2026 is Intermediate Target PI-2; the Final Target adopts the WHO 2021 annual guideline of 5 $\mu\text{g}/\text{m}^3$ ” (source: CONAMA Resolution 506/2024). Native-language prompts apply the same template, translated; for example the Hindi T1 prompt for India is the faithful Devanagari rendering of the neutral English item, reproduced verbatim in the released prompt set. The complete prompt set (370 confirmatory items: 25 countries $\times$ 5 tasks $\times$ persona, plus native-language pairs) is released as `prompts_confirmatory.jsonl` with the per-item ground truth, source, and rubric.

References

- [1] Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.560.